Cassini ISS F Ring Mosaics User Guide

V1.0

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1 Introduction

Saturn’s F ring is one of the most dynamic and scientifically intriguing features in the solar system. Discovered in 1979 by the Pioneer 11 spacecraft, the F ring is a narrow, complex, and ever-changing structure located just outside Saturn’s main ring system. Its unique morphology, characterized by multiple strands, clumps, and transient features, along with the influence of neighboring moons Prometheus and Pandora, has made it a focal point for studies of planetary ring dynamics, satellite-ring interactions, and the processes governing the evolution of ring systems.

Before 2004, quality images of the F ring were limited to a handful produced by the Voyager 1 and 2 spacecraft during their short fly-bys in 1980 and 1981. This all changed with the arrival of the Cassini Orbiter at Saturn on June 30, 2004. During its subsequent 13 years at Saturn, Cassini used its Imaging Science Subsystem (ISS) (Knowles 2018; Porco et al. 2004) to observe the F ring more than 100,000 times, providing a treasure trove of data that is unlikely to be surpassed for generations.

This *User Guide* provides a detailed description of the organization and contents of the Cassini ISS F Ring Mosaics bundle archived at NASA’s Planetary Data System Ring-Moon Systems Node under NASA CDAP grant 80NSSC21K0527: “The recent history of Saturn's dusty rings”. The purpose of this archive is to provide a carefully curated and documented set of high-quality images of the F ring taken by the ISS Narrow Angle Camera (NAC) and Wide Angle Camera (WAC) (Porco et al. 2004) using clear filters. Special emphasis is placed on “FMOVIEs”, sequences of dozens to hundreds of observations taken in rapid succession at a single inertial longitude while the F ring orbited through the field of view. Such movies provide a detailed picture of the F ring through most or all of a complete orbit.

All relevant images were calibrated using CISSCAL 4.0 (Knowles 2018) and then reprojected onto a grid with radius relative to the F ring core on the vertical axis and co-rotating longitude on the horizontal axis (see Section 2.4). Multiple images are stitched together to form more complete mosaics. Images and mosaics are both organized by Cassini “observation name” (see Section 2.1).

The archive includes the reprojected images, mosaics, and mosaics that have been adjusted by subtracting a background model. Each image and mosaic is supplemented by a table containing metadata such as resolution, phase angle, and emission angle for each valid co-rotating longitude. Reprojected images have an additional file describing the spacecraft pointing used to perform the reprojection. Mosaics have an additional file listing the reprojected images that were used to create the mosaic. A series of browse images in various sizes is provided for each image as well. In total 20,303 reprojected images are provided along with 302 associated mosaics.

First time users are encouraged to read or skim this entire *User Guide* before making use of the archived results.

2 Image Selection and Processing

2.1 Image Selection

Cassini observations were organized into observing campaigns. Each campaign was given a unique identifier (the observation name), which has the general format:

[INST]_[REV][TI]_[UNIQUENAME]_[PINST]

where INST is the abbreviation for the instrument being commanded (always “ISS” here), REV is the rev (orbit) number, TI is a 2-character target identifier, UNIQUENAME is a unique descriptor chosen by the observation creator, and PINST indicates the name of the prime instrument if other than the ISS

(Knowles 2018). Many of these campaigns were specifically designed to observe the rings in ways that would include the F ring:

- AZSCAN: Azimuthal scan around the rings.
- EGAPMOVMP: Movie of the Encke Gap at moderate phase angle.
- FxxPHASE: Observe the F ring at xx degrees phase angle.
- FMOVIE: A stare in a fixed orientation to image F ring material passing through the NAC field of view.
- FRSTRCHAN: Prometheus & F ring streamer-channel movie. NAC tracks Prometheus to follow the evolution of the streamer-channel feature raised in the F ring by gravitational interaction with Prometheus at the satellite's apoapse passage.
- PROPRETRG: Re-targeting known "propeller" moons to track their orbits.
- SPK: Basically, any observation that starts with "SPK" is a spoke obs. They tend to have suffixes appended to them. Common ones are DF/LF for Dark Face/Lit Face, HP/LP for High Phase/Low Phase, and LR/MR/HR for Low Resolution/Medium Resolution/High Resolution.
- SPKMV: Spoke periodicity movies. These are 'point and stare' observations where we select a ring ansa and image at regular intervals (usually with the WAC) to watch spokes move through the field of view.

Each such observation was manually inspected to determine the presence of the F ring. In addition, the complete set of ~100,000 images of the F ring was manually inspected for other high-quality images of the F ring that were not categorized into one of the above observation types.

In general, an image was included in our data set if:

1. It was a member of a series of images taken at different co-rotating longitudes that could be stitched together into a mosaic;
2. It was taken with the "clear" filters;
3. It was not corrupted or partially downloaded; and
4. It had a minimum radial resolution for the F ring of at most ~80 km/pixel.

Some images were included that do not meet requirement #1. For example, if a single WAC image contained enough of the F ring to make a good mosaic, it was included. Also, some image sequences were included where the camera followed a single co-rotating longitude at different inertial longitudes. In addition, some images were included that do not meet requirement #4. These included 42 "SPOKEMOV" sequences taken at relatively low resolution by the WAC. While the reprojected images for these sequences are not of very high quality, the resulting mosaics are nevertheless sufficient for research, such as photometry, that does not require high-resolution analysis. In total, 20,303 images were chosen for inclusion.

Image sequences were grouped under their original observation name converted to lower case. In some cases, a single observation contained multiple distinct image sequences that would make different mosaics. In these cases, a unique numeric suffix was appended to the observation name.

Ultimately the following classes of observations were selected. Each is indicated in the data archive by a comment in the mosaic PDS4 label and also by a note in the index file.

2.1.1 Standard inertial F movies

A standard "F movie" consists of a series of images taken while Cassini stared at approximately the same inertial longitude. The F ring rotated under the camera for some or all of a complete orbit, providing a partial or complete view of the entire longitudinal extent of the ring. Such observations are considered the default and are thus not marked with a special note.

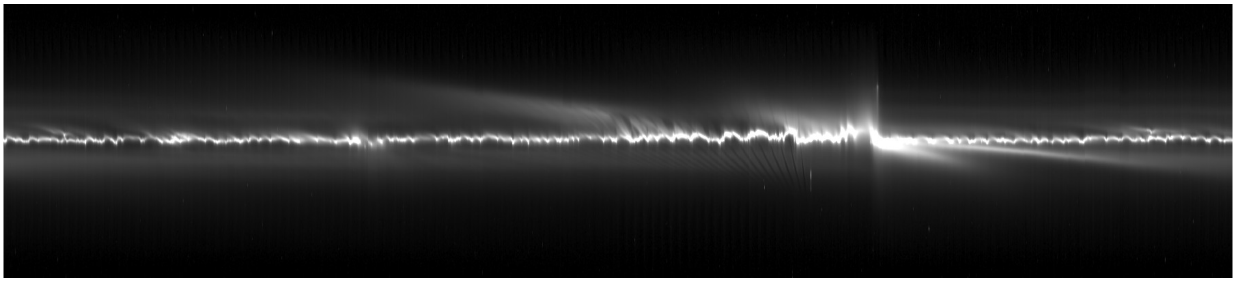

Figure 1: Mosaic of observation iss_039rf_fmovie001_vims, a standard "F movie".

It is important to realize that it was the F ring that was rotating, not the camera that was instantaneously observing different parts of the ring. As a result, the resulting image sequence and mosaic does not actually provide a view of the F ring around its entire orbit. Instead, it provides repeated views of the F ring at a single inertial longitude, and thus at a single point in its eccentric orbit. This can affect the interpretation of morphological features such as streamer-channels, which change shape throughout the orbit.

An example of a standard mosaic is shown in Figure 1.

2.1.2 Mx: Observations that are broken into chunks

Although Cassini observing campaigns are indicated by their observation name, sometimes a single campaign produced data that was appropriate for more than one mosaic. In these cases, we produced multiple mosaics named after the observation name with a numerical suffix. The notes field in the index file includes "M1", "M2", "M3", or "M4", indicating how the different chunks are related to each other.

2.1.2.1 M1: Multiple contiguous observations of the same inertial longitude range

In some cases, an observation observed a single inertial longitude for more than a complete orbit of the F ring. In these cases, the observation was broken into multiple chunks and each complete or partial final orbit of the F ring was used to create a separate mosaic. Such observations are noted in the index file with "M1". The following mosaics have this flag:

- iss_198ri_spokemov005_prime_1/2
- iss_199ri_spokemov002_prime_1/2
- iss_199ri_spokemov003_prime_1/2/3
- iss_199ri_spokemov004_prime_1/2
- iss_199ri_spokemov006_prime_1/2/3
- iss_200ri_spokemov004_prime_1/2/3
- iss_201ri_spokemov001_prime_1/2
- iss_201ri_spokemov009_prime_1/2

2.1.2.2 M2: Pairs of observations taken at inertial longitudes 180 degrees apart

Some observations were performed in pairs, where a particular range of co-rotating longitudes was observed at one inertial longitude, and then approximately the same range of co-rotating longitudes was observed at the inertial longitude roughly 180 degrees away. Such pairs provide insight into how various portions of the F ring are on different orbits. These observations are divided into two chunks and noted with "M2". The following mosaics have this flag:

- iss_085rf_fmovie003_prime_1/2
- iss_094rf_fmovie001_prime_1/2
- iss_112rf_fmovie002_prime_1/2

- iss_173rf_fmovie001_prime_1/2
- iss_174rf_frstrchan001_prime_1/2
- iss_177rf_fmovie001_prime_1/2
- iss_177rf_frstrchan001_prime_1/2
- iss_183rf_fmovie001_prime_1/2
- iss_193rf_fmovie001_prime_1/2
- iss_194rf_fmovie001_prime_1/2
- iss_198rf_fmovie001_prime_1/2
- iss_201rf_fmovie001_prime_1/2
- iss_202rf_fmovie001_prime_1/2
- iss_203rf_fmovie001_prime_1/2
- iss_208rf_fmovie001_prime_1/2
- iss_208rf_fmovie002_prime_1/2
- iss_209rf_fmovie001_prime_1/2
- iss_212rf_fmovie001_prime_1/2
- iss_232ri_egapmovmp001_prime_1/2
- iss_233rf_fmovie001_prime_1/2
- iss_235rf_fmovie002_prime_1/2
- iss_239rf_fmovie002_prime_1/2
- iss_243rf_fmovie001_prime_1/2
- iss_256rf_fmovie001_prime_1/2
- iss_276ri_hiresafrg001_prime_1/2

An example of such a pair of mosaics is shown in Figure 2.

2.1.2.3 M3: Multiple observations of the same co-rotating longitude range but different inertial longitude ranges

While standard “F movies” are taken by staring at a single inertial longitude in isolation, some observations were performed by taking multiple “F movies” in a row, each at a different inertial longitude. These observations are marked with “M3”. The following mosaics have this flag:

- iss_105ri_tmapn45lp001_cirs_1
- iss_105ri_tmapn45lp001_cirs_2
- iss_105ri_tmapn45lp001_cirs_3
- iss_105ri_tmapn45lp001_cirs_4
- iss_105ri_tmapn45lp001_cirs_5
- iss_105ri_tmapn45lp001_cirs_6
- iss_262rf_fmovie001_prime_01−17
- iss_268rf_fmovie001_prime_2−9

An example is shown in Figure 3.

2.1.2.4 M4: Observations of different co-rotating and different inertial longitudes

Finally, some observations follow neither a single co-rotating longitude nor a single inertial longitude. This is often the case for occultations. If both the ingress and egress occultations are imaged, the stellar position in the two has no relation to either the co-rotating or inertial longitude. There are also some observations that are broken into different phases similar to “M3”, but each chunk observed both

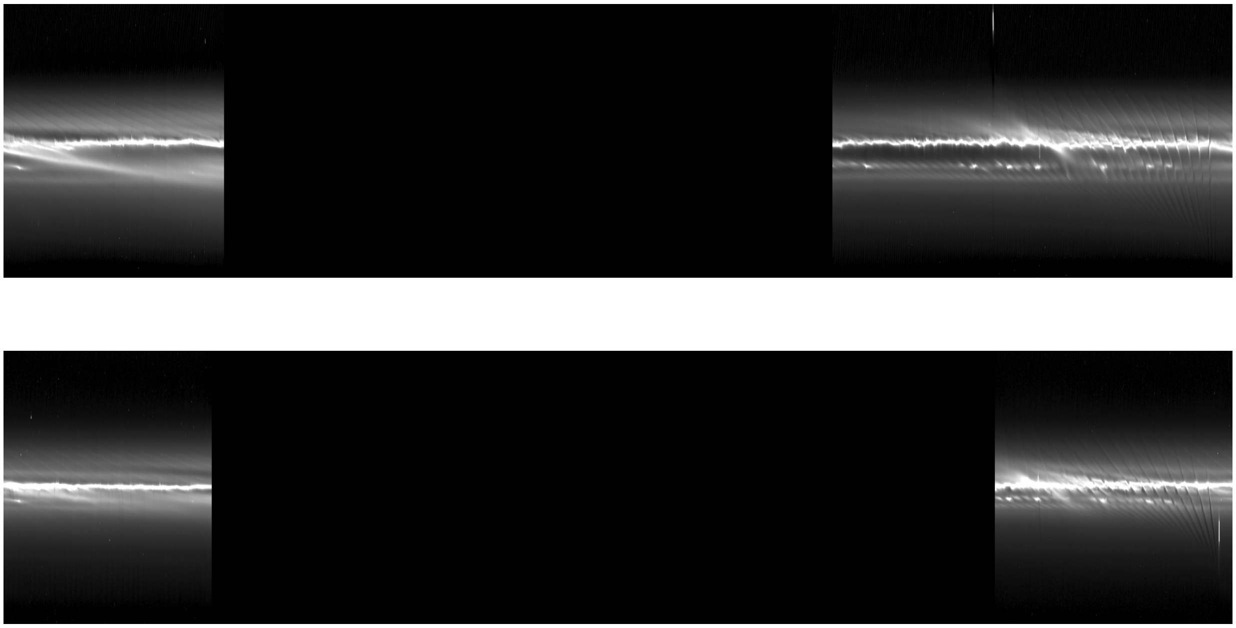

Figure 2: Mosaics of observations iss_112rf_fmovie002_prime_1 (top) and iss_112rf_fmovie002_prime_2 (bottom), taken at inertial longitudes roughly 180 degrees apart.

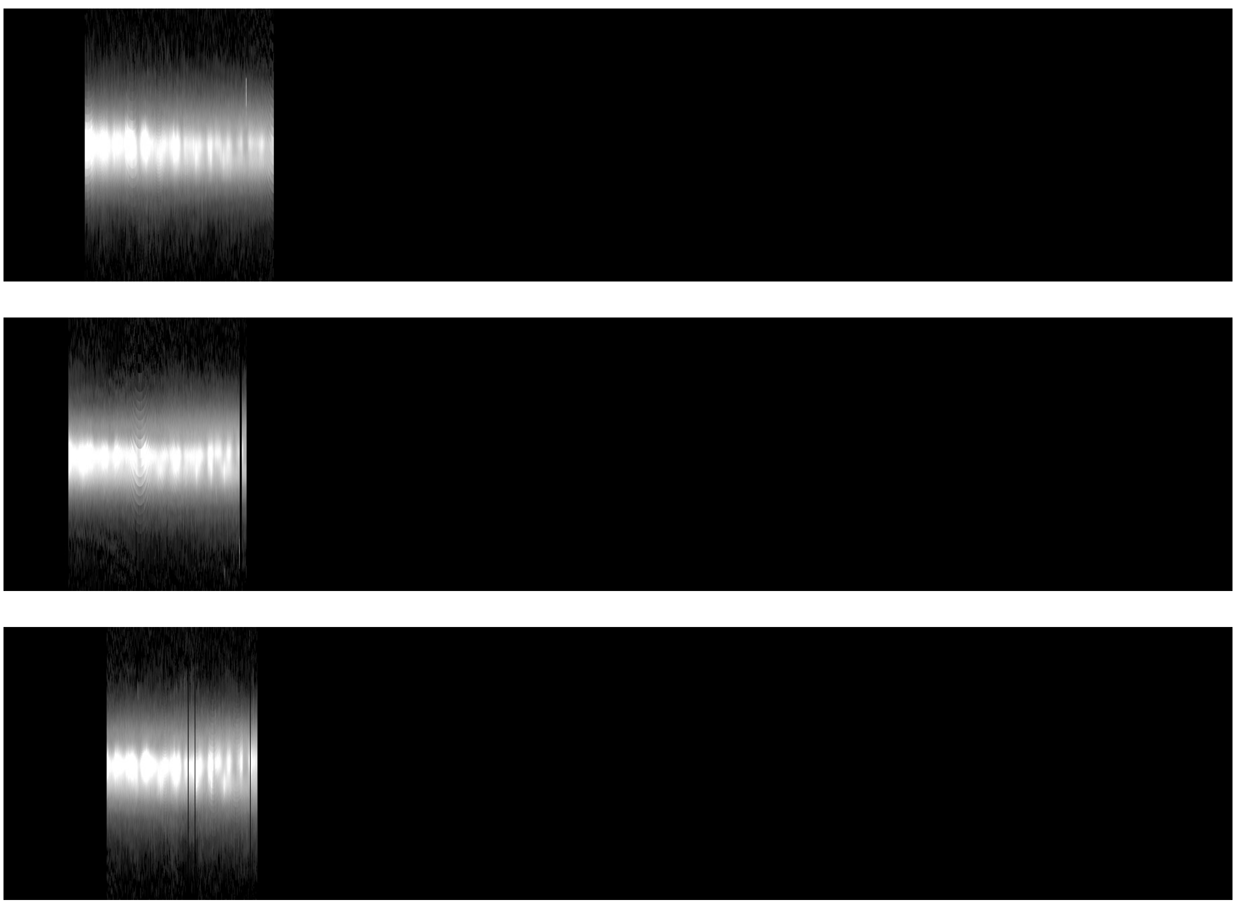

Figure 3: A series of observations covering the same co-rotating longitudes but taken at progressive inertial longitudes. iss_105ri_tmapn45lp001_cirs_1 (top) was taken at 78°–130°; iss_105ri_tmapn45lp001_cirs_2 (middle) was taken at 109°–154°; iss_105ri_tmapn45lp001_cirs_3 (bottom) was taken at 136°–185°.

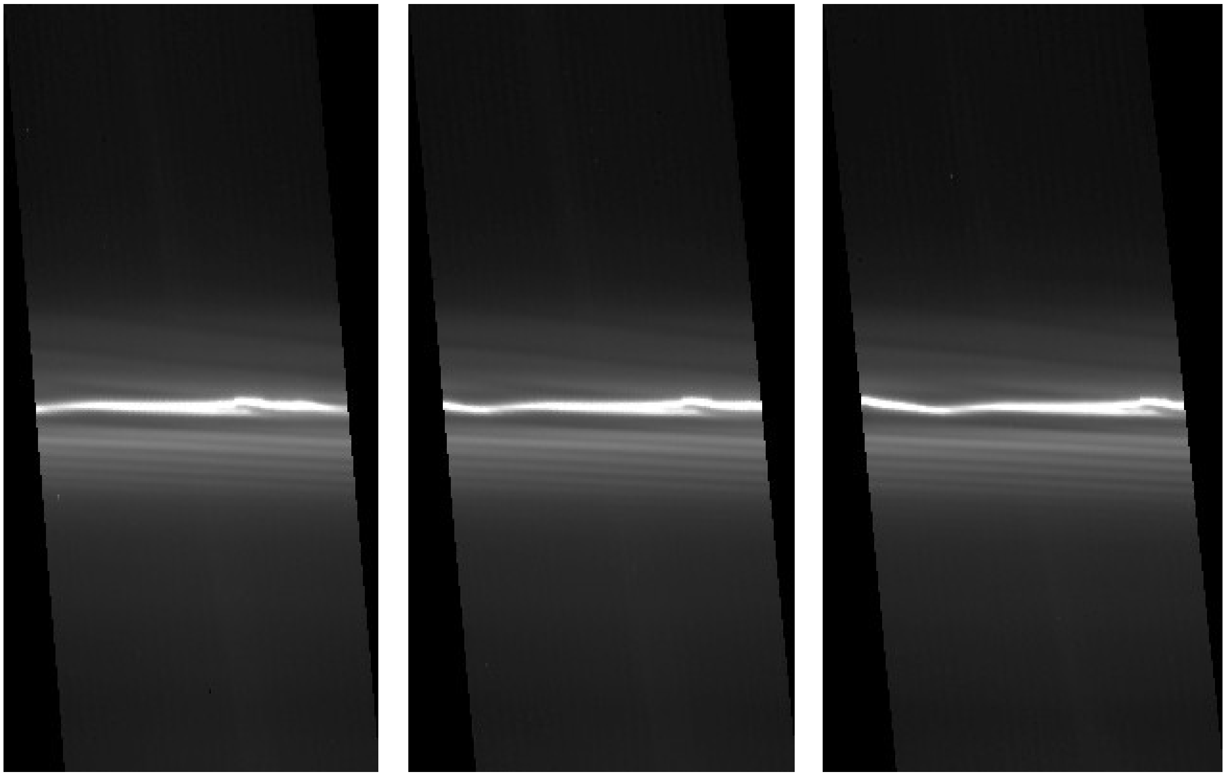

Figure 4: Three reprojected images from mosaic iss_191rf_fmovie001_prime_1 taken at approximately the same 3° of co-rotating longitudes. Image #1, 1748745661n (left), was taken at inertial longitudes 6.0°–9.2°; image #5, 1748746261n (middle), was taken at inertial longitudes 9.6°–12.9°; image #9, 1748746865n (right), was taken at inertial longitudes 13.3°–16.6°.

different co-rotating and inertial longitudes. These observations are marked with "M4". The following mosaics have this flag:

- iss_185ri_rhyaocc001_vims_1/2
- iss_191rf_fmovie001_prime_1/2/3/4/5
- iss_201ri_l2pupocc001_vims_1/2
- iss_206ri_propretrg001_prime_1/2/3
- iss_239ri_hiresafrg001_prime_1/2
- iss_253rf_fmovie001_prime_1/2/3
- iss_268rf_fmovie001_prime_1 (for this mosaic, all of the other chunks are marked with "M3", but this chunk behaves differently)

2.1.3 R: Follows one co-rotating longitude range with different inertial longitudes

A small number of observations follow a single co-rotating longitude, taking one image at each inertial longitude. These are similar to the "M3" observations above, but contain such a small range of co-rotating longitudes in each image that it is not worth breaking them into separate mosaics. In these cases, a single mosaic is produced, but because the images overlap the mosaic is not particularly useful. It is recommended instead to look at the corresponding reprojected images. These observations are marked with "R". If a single observation is split into chunks, "R" may be paired with an appropriate "M" note. An example is shown in Figure 4.

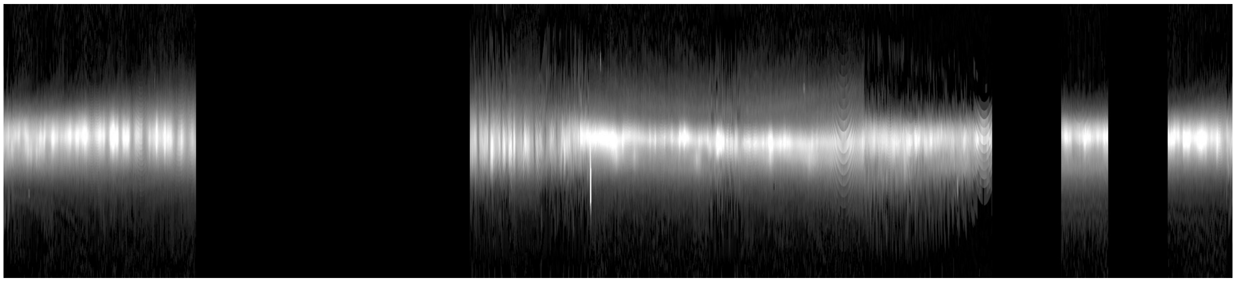

Figure 5: Non-inertial mosaic iss_098ri_tmapn30lp001_cirs, which covers most of the full co-rotating range 0–359.98° taken from inertial longitudes 93-309°.

2.1.4 N: Non-inertial

In a few cases, especially those involving wide-field views taken by the WAC instrument, an observation can't be classified as either inertial or non-inertial because the camera doesn't actually follow anything, it just takes a single or short series of images that cover large portions of the F ring. These cases are marked with "N". An example is shown in Figure 5.

2.1.5 O: Occultation

Several observations were designed to monitor a stellar occultation of the F ring core. These observations continuously observed the position of the star as the F ring rotated. Since each image has the star in it, the mosaic may show the star multiple times. Like the M3 mosaics, these mosaics are not particularly useful by themselves and it is recommended to look at the original reprojected images instead. Also, since these are "ride-along" observations where the Cassini Visible and Infrared Mapping Spectrometer (VIMS) or Ultraviolet Imaging Spectrograph (UVIS) instruments were the primary instrument measuring the photon flux, the ISS image sequences do not necessarily show the moment when the star was occulted by the F ring.

The observations ISS_185RI_RHYAOCC001_VIMS and ISS_201RI_L2PUPOCC001_VIMS observed both the ingress and egress and thus are split into two chunks and marked with "M4".

The occultation observations are marked with "O". The following mosaics have this flag.

- iss_172ri_betpegocc001_vims
- iss_172st_urgampeg001_uvis
- iss_180ri_rlyrocc001_vims
- iss_180ri_rcasocc001_vims
- iss_185ri_rhyaocc001_vims_1/2
- iss_194ri_mucepocc001_vims
- iss_196ri_betandocc001_vims
- iss_197ri_whyaocc001_vims
- iss_198ri_rlyrocc001_vims
- iss_201ri_l2pupocc001_vims_1/2
- iss_205ri_l2pupocc002_vims
- iss_206ri_l2pupocc002_vims

2.2 Calibration

Each image was calibrated using the CISSCAL 4.0 pipeline (Knowles 2018). The creation of calibrated images had already been performed by the PDS Ring-Moon Systems Node, and the calibrated images were retrieved from that archive.[1] The calibrated files at the time of writing are being migrated to the PDS4 standard, and those will be archived as collections within the *cassini_iss_cruise* and *cassini_iss_saturn* bundles.[2]

2.3 Navigation

The spacecraft pointing information present in SPICE C kernels is known to have errors on the order of dozens of NAC pixels. Before processing each image, an automatic navigation procedure (French et al. 2014b, 2016) was performed that matched the image with features that were predicted to be present in the image such as stars and the edge of the A ring. Once a mosaic was created from a set of images, the quality of the navigation was evaluated by eye and, in some cases, the navigation of some images was manually adjusted for the best visual result. The navigation adjustment for each image is reported in a supplemental file in the **data_reproj_img** collection. See Section 3.2.1.

2.4 Reprojection

Each calibrated image was converted to a reprojected image by mapping the pixels in the source image onto a two-dimensional grid with co-rotating longitude on the X axis and distance from the nominal F ring core (Δr) on the Y axis.

We use the orbit fit from (Albers et al. 2012) Table 3 fit #2:

n = 581.964 °/day
a = 140,221.3 km
e = 0.00235
ϖ_0 = 24.2°
$\dot{\varpi}$ = 2.70025 °/day
Ω_0 = 15.0°

Co-rotating longitude is based on a reference frame that rotates with the mean motion of the F ring. We use 2007-01-01T00:00:00Z as the epoch when co-rotating longitude and inertial longitude are equal. To convert from inertial to co-rotating longitude:

$$corot = (inertial - n * (ET - epoch\_ET) / 86400) \% 360$$

and to convert from co-rotating to inertial longitude:

$$inertial = (corot + n * (ET - epoch\_ET) / 86400) \% 360$$

where *ET* is the ephemeris (SPICE) time of the image in seconds and *epoch_ET* is the co-rotating epoch in seconds.

Note that the choice of epoch is arbitrary and when comparing absolute co-rotating longitudes from different sources it is important to convert to the new reference frame. For example, Murray et al. (2008) used 2007-01-01T12:00:00Z, French et al. (2012) used 2004-01-01T12:00:00Z, and French et al. (2014a) used 2007-01-01T00:00:00Z.

To determine the radius of the F ring core, we use the standard formula:

[1] https://pds-rings.seti.org/holdings/calibrated/COISS_2xxx/
[2] https://pds-rings.seti.org/pds4/bundles/cassini_iss

$$r = \frac{a(1-e^2)}{1 + e\cos\left(\varpi_0 + \dot{\varpi}\frac{ET}{86400}\right)}$$

The reprojection grid runs from 0° to 359.98° in steps of 0.02° for co-rotating longitude (18,000 pixels), and −1000 km to +1000 km in steps of 5 km for Δr (401 pixels). For each co-rotating longitude, the radius is relative to the absolute radius of the core at the associated inertial longitude and time. This means that, while the core of the F ring varies in radius from 139,892 km to 140,551 km throughout its orbit, the core in the reprojected image will always look like an approximately straight line at $\Delta r = 0$, corresponding to Y = 200.

As the majority of reprojected images cover only a few degrees of co-rotating longitude, archiving files of the complete 18,000 × 401 grid would be extremely wasteful. As such, the reprojected image data products are clipped on the X axis to the minimum and maximum co-rotating longitudes that contain valid data; see the associated label for information on these values. In the rare case where the extent of a reprojected image wraps around from 358.98° to 0°, the full grid size is stored. In all cases, if there is missing data within the remaining grid (either due to insufficient radial extent available in the source image, or discontinuous co-rotating longitudes), the missing data is marked with the sentinel value –999.

During the reprojection process, several other parameters are computed for each pixel:

- Phase angle (°)
- Emission angle (°)
- Radial resolution (km/pixel)
- Longitudinal resolution (°/pixel)

For each co-rotating longitude, the mean of each parameter is computed and reported in a table that is associated with each reprojected image. Incidence angle is also computed, but because it does not change over the image the single value is reported in the label.

2.5 Mosaic Creation

Mosaics were created by stitching together one or more reprojected images. The resulting grid for a mosaic is the same 18,000 × 401 pixel grid as used for each image. For each co-rotating longitude in the mosaic, the entire radial stripe is taken from a single image. The image is chosen first based on which image has the most valid pixels at that co-rotating longitude; when there is a tie, the image with the smallest mean radial resolution is used. This guarantees that the resulting mosaic will have the most data and the highest resolution possible.

Similar to reprojected images, for each co-rotating longitude several parameters are stored and provided in a table associated with each mosaic:

- Mean phase angle (°)
- Mean emission angle (°)
- Mean radial resolution (km/pixel)
- Mean longitudinal resolution (°/pixel)
- Image number

Incidence angle is assumed to be the same for all reprojected images in a mosaic and is reported in the label.

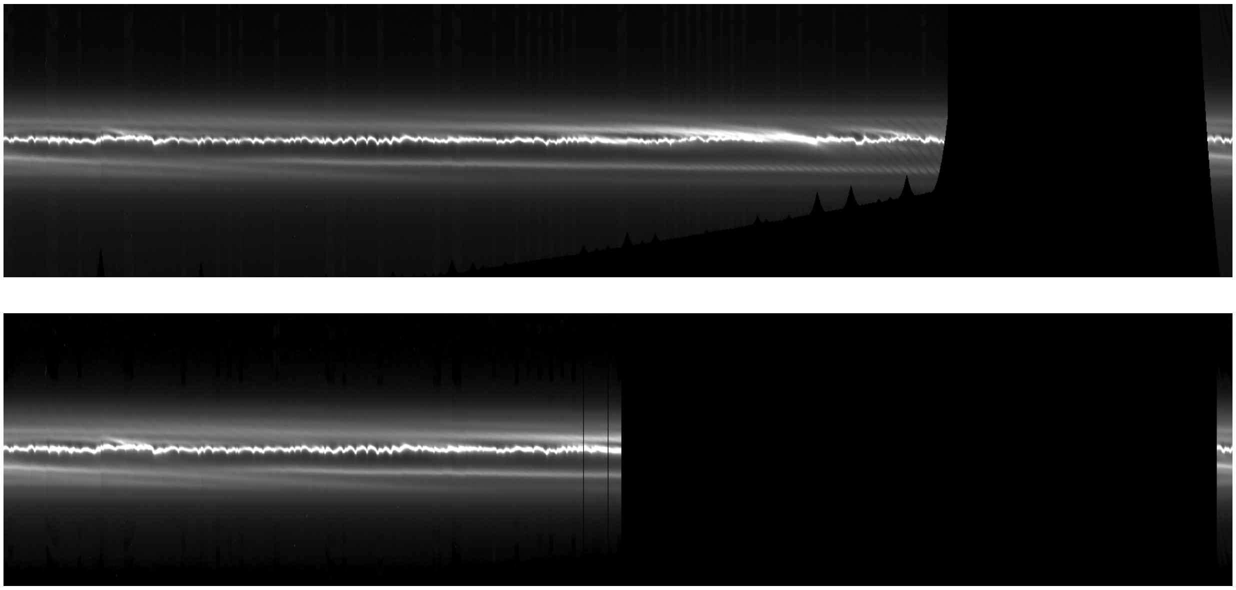

Figure 6: Original mosaic iss_007ri_hpmrdfmov001_prime (top) and associated background-subtracted mosaic (bottom). The lack of some data in the original mosaic prevents the creation of a background model for those longitudes, resulting in a background-subtracted mosaic with substantially fewer valid longitudes.

2.6 Background Modeling

The final step in the data production pipeline is modeling and subtracting the background gradient for each mosaic. The background gradient is modeled separately for each co-rotating longitude. A region normally consisting of the innermost 50 pixels ($\Delta r = -1000$ to -750 km) and the outermost 50 pixels ($\Delta r = +750$ to $+1000$ km) is used to fit a linear model which is then subtracted from the complete 401 pixel radial slice. When creating the model, pixel values that are statistically anomalous compared to nearby longitudes are automatically masked. This removes stars, Prometheus and Pandora, and bad pixels from the background model computation. In some cases, the standard range of 50 pixels on each side is not appropriate. Reasons include large objects such as Prometheus covering the entire 50 pixel range or the reprojected source images not providing data for the entire radial range (which would require the background margins to be increased in size) and very low resolution images where the F ring bleeds over into the background region (which would require background margins to be reduced). Such determinations were made through careful examination of each mosaic and the extent of the background margin was manually adjusted as needed. Whenever possible the new margins were chosen to have minimal effect on computation of equivalent width. The chosen pixel limits are reported in the label for the background-subtracted mosaic.

In cases where, even after adjusting the pixel limits, there was insufficient radial data to create a background model for some longitudes, those longitudes were marked as invalid and deleted from the background-subtracted mosaic. As a result, the background-subtracted mosaic may have fewer (and sometimes substantially fewer) valid longitudes than the original mosaic. An example is shown in Figure 6.

3 Bundle Organization and Directory Structure

The Cassini ISS F Ring Mosaics bundle is organized in a hierarchical directory structure, with each directory containing a PDS4 collection of a different type of derived data product.[3] In addition, there are directories for documentation, context, SPICE kernels, and XML schema. The top-level directory structure is:

```
cassini_iss_fring_mosaics_rsfrench2025/
├── bundle.lblx
├── readme.txt
├── browse_mosaic/
├── browse_mosaic_bkg_sub/
├── browse_reproj_img/
├── context/
├── data_mosaic/
├── data_mosaic_bkg_sub/
├── data_reproj_img/
├── document/
├── spice_kernels/
└── xml_schema/
```

Each directory is described in the sections below. The *context* and *xml_schema* directories are used internally by PDS and will not be discussed further.

3.1 The Top-Level Directory

The top-level directory contains the following files:

- **bundle.lblx**: The PDS4 bundle label, providing the root metadata for the entire archive.
- **readme.txt**: A text file directing the user to this *User Guide*.

3.2 The *browse* and *data* directories

For each derived data type (reprojected image, mosaic, and background-subtracted mosaic) there is a **browse** and a **data** directory. These directories have the same internal structure:

```
<dirname>/
├── collection_<dirname>.csv
├── collection_<dirname>.lblx
├── OBSID1/
├── OBSID2/
├── ...
└── OBSIDn/
```

The **collection_<dirname>.csv** file contains a list of all of the data products present for all OBSIDs in this directory, and the accompanying label describes its metadata. The data products themselves are separated by OBSID (including any numeric suffix as discussed earlier) for ease of retrieval and to keep the number of files in each subdirectory relatively small.

Each derived data type has its own file structure within each OBSID directory.

3.2.1 data_reproj_img

This directory contains the reprojected images organized by OBSID. The directory structure is:

[3] Refer to the documents in Section 4.1 for further information on the PDS4 concepts of bundle, collection, and product.

```
data_reproj_img/
├── collection_data_reproj_img.csv
├── collection_data_reproj_img.lblx
├── OBSID1/
│   ├── IMGID1_reproj_img.lblx
│   ├── IMGID1_reproj_img.img
│   ├── IMGID1_reproj_img_suppl.txt
│   ├── IMGID1_reproj_img_metadata_params.tab
│   └── ...
└── ...
```

The file **IMGID_reproj_img.lblx** contains all of the details about the reprojected image including the reprojection and viewing geometry:


```
<rings:epoch_reprojection_basis_utc>2007-01-01T00:00:00.000Z</rings:epoch_reprojection_basis_utc>
<rings:reprojection_plane>Equator</rings:reprojection_plane>
<rings:corotating_flag>Y</rings:corotating_flag>
<rings:corotation_rate unit="deg/s">0.006735694444</rings:corotation_rate> <!-- 581.964 deg/day -->
<rings:mean_phase_angle unit="deg">36.119</rings:mean_phase_angle>
<rings:minimum_phase_angle unit="deg">35.990</rings:minimum_phase_angle>
<rings:maximum_phase_angle unit="deg">36.182</rings:maximum_phase_angle>
<rings:mean_emission_angle unit="deg">97.421</rings:mean_emission_angle>
<rings:minimum_emission_angle unit="deg">97.200</rings:minimum_emission_angle>
<rings:maximum_emission_angle unit="deg">97.651</rings:maximum_emission_angle>
<rings:mean_incidence_angle unit="deg">88.821387</rings:mean_incidence_angle>
<rings:minimum_incidence_angle unit="deg">88.821387</rings:minimum_incidence_angle>
<rings:maximum_incidence_angle unit="deg">88.821387</rings:maximum_incidence_angle>
<rings:minimum_corotating_ring_longitude unit="deg">286.14</rings:minimum_corotating_ring_longitude>
<rings:maximum_corotating_ring_longitude unit="deg">308.60</rings:maximum_corotating_ring_longitude>
<rings:minimum_ring_radius unit="km">138894.242</rings:minimum_ring_radius>
<rings:maximum_ring_radius unit="km">140934.268</rings:maximum_ring_radius>
<rings:Reprojection_Grid_Parameters>
  <local_identifier>reproj_grid</local_identifier>
  <rings:reprojection_grid_radial_sampling_interval unit="km">
    5.0</rings:reprojection_grid_radial_sampling_interval>
  <rings:reprojection_grid_longitudinal_sampling_interval unit="deg">
    0.02</rings:reprojection_grid_longitudinal_sampling_interval>
  <rings:minimum_radial_resolution unit="km">5.390</rings:minimum_radial_resolution>
  <rings:maximum_radial_resolution unit="km">9.827</rings:maximum_radial_resolution>
  <rings:mean_radial_resolution unit="km">7.001</rings:mean_radial_resolution>
  <rings:minimum_longitudinal_resolution unit="deg">0.016</rings:minimum_longitudinal_resolution>
  <rings:maximum_longitudinal_resolution unit="deg">0.018</rings:maximum_longitudinal_resolution>
  <rings:mean_longitudinal_resolution unit="deg">0.017</rings:mean_longitudinal_resolution>
</rings:Reprojection_Grid_Parameters>
```


and the details of the image file, **IMGID_reproj_img.img**:


```
<File_Area_Observational>
  <File>
    <file_name>1622049830n_reproj_img.img</file_name>
    <creation_date_time>2025-07-25T22:48:10Z</creation_date_time>
    <file_size unit="byte">1050620</file_size>
    <md5_checksum>ad24ba420a8705285effa703a950b61e</md5_checksum>
  </File>
  <!-- The calibrated, reprojected single image -->
  <Array_2D_Image>
    <local_identifier>image</local_identifier>
    <offset unit="byte">0</offset>
    <axes>2</axes>
    <axis_index_order>Last Index Fastest</axis_index_order>
    <Element_Array>
      <data_type>IEEE754LSBSingle</data_type>
      <unit>I/F</unit>
    </Element_Array>
    <Axis_Array>
      <axis_name>Line</axis_name> <!-- vertical (radial) -->
      <elements>401</elements>
      <sequence_number>1</sequence_number>
    </Axis_Array>
```

```
        <Axis_Array>
            <axis_name>Sample</axis_name> <!-- horizontal (longitude) -->
            <elements>655</elements>
            <sequence_number>2</sequence_number>
        </Axis_Array>
        <Special_Constants>
            <!-- Data was missing or corrupted in the original image (either because it was off
                the edge of the FOV or because of a transmission error resulting in a partial or
                corrupted image) -->
            <missing_constant>-999</missing_constant>
        </Special_Constants>
    </Array_2D_Image>
</File_Area_Observational>
```


The image file is a pure binary file with no headers or record separators. Data is stored in LSB IEEE754 single-precision (32-bit) floating point format. The Y axis (Δr) always has size 401. The X axis (co-rotating longitude) has a variable size depending on the number of longitudes that contain valid data (see Section 2.4). Any array elements that do not contain valid data will be marked with the value –999.

An image file can be read easily in Python using NumPy. Because the Y axis always has size 401, the "–1" argument to **np.reshape** can be used to infer the size of the X axis. There are many ways to handle the invalid array elements, but one simple way is to used NumPy's MaskedArray feature.

```
Import numpy as np
Import numpy.ma as ma
raw_data = np.fromfile('1622033938n_reproj_img.img', dtype=np.float32).reshape((401, -1))
masked_data = ma.masked_equal(x.view(ma.MaskedArray), -999)
```

An associated metadata file, **IMGID1_reproj_img_metadata_params.tab**, contains information about each valid longitude. This table has the general format:

```
Corotating Longitude, Inertial Longitude, Radial Resolution, Angular Resolution, Phase Angle, Emission Angle
286.14, 338.569,   9.82725,   0.00027,  36.093438,  97.650969
286.16, 338.589,   9.81850,   0.00027,  36.093433,  97.650595
286.18, 338.609,   9.80948,   0.00027,  36.093351,  97.650209
286.20, 338.629,   9.79825,   0.00027,  36.093280,  97.649716
286.22, 338.649,   9.78796,   0.00027,  36.093194,  97.649278
286.24, 338.669,   9.77744,   0.00027,  36.093139,  97.648815
286.26, 338.689,   9.76799,   0.00027,  36.093132,  97.648411
[…]
```

where each line refers, in sequence, to an X index in the image array. This table gives the actual co-rotating longitude that is defined by that index as well as additional useful information such as the inertial longitude, radial resolution, angular (longitudinal) resolution, phase angle, and emission angle.

Finally, a supplemental information file, **IMGID1_reproj_img_suppl.txt**, describes the navigation process and the derived pointing for the image (see Section 2.3):

```
Source Data Product ID = 1736795703n_calib
Image Start Time (SCLK) = 1/1736795701.243
Image Start Time (UTC) = 2013-01-13T18:20:56.381Z
Image Mid Time (SCLK) = 1/1736795702.180
Image Mid Time (UTC) = 2013-01-13T18:20:57.131Z
Image Stop Time (SCLK) = 1/1736795703.118
Image Stop Time (UTC) = 2013-01-13T18:20:57.881Z
Trajectory Kernels Query Time = Observation mid time
Stellar Aberration Correction = No
Light Travel Time Correction = No
Navigation Type = Ring and/or Satellite Models
Navigated Boresight RA = 287.227 deg (19h08m54.446s)
Navigated Boresight Dec = 46.7338 deg (+046d44m1.850s)
Navigated Boresight Roll = 22.6952 deg
C-Matrix =
```

```
 0.7979776914   0.5415732280   0.2644428914
-0.5674768892   0.5273925427   0.6323188168
 0.2029817339  -0.6546415390   0.7281777744
```

3.2.2 browse_reproj_img

This directory contains browse images for each reprojected image. The directory structure is:

```
browse_reproj_img/
├── collection_browse_reproj_img.csv
├── collection_browse_reproj_img.lblx
├── OBSID1/
│   ├── IMGID1_browse_reproj_img.lblx
│   ├── IMGID1_browse_reproj_img_full.png
│   ├── IMGID1_browse_reproj_img_med.png
│   ├── IMGID1_browse_reproj_img_small.png
│   ├── IMGID1_browse_reproj_img_thumb.png
│   └── …
└── …
```

Four sizes of browse images are provided. "Thumb" images are always 100×100 pixels and are downsampled and padded as necessary from the reprojected image. "Small" images are always 200×200 pixels and are similarly downsampled and padded. "Med" images are downsampled by 10 in the X dimension and are always 401 pixels in the Y dimension, thus having a maximum size of 1800×401. "Full" images are always the same size as the reprojected image and guaranteed to be 401 pixels in the Y dimension. All image types are contrast-stretched to use the full available greyscale. Missing data in the reprojected image is indicated by black.

3.2.3 data_mosaic and *data_mosaic_bkg_sub*

These directories have identical structure and contain the original and background-subtracted mosaics, respectively. The directory structures are:

```
data_mosaic/
├── collection_data_mosaic.csv
├── collection_data_mosaic.lblx
├── OBSID1/
│   ├── OBSID1_mosaic.lblx
│   ├── OBSID1_mosaic.img
│   ├── OBSID1_mosaic_metadata_params.tab
│   ├── OBSID1_mosaic_metadata_src_imgs.tab
│   └── …
└── …
```

and:

```
data_mosaic_bkg_sub/
├── collection_data_mosaic_bkg_sub.csv
├── collection_data_mosaic_bkg_sub.lblx
├── OBSID1/
│   ├── OBSID1_mosaic_bkg_sub.lblx
│   ├── OBSID1_mosaic_bkg_sub.img
│   ├── OBSID1_mosaic_bkg_sub_metadata_params.tab
│   ├── OBSID1_mosaic_bkg_sub_metadata_src_imgs.tab
│   └── …
└── …
```

The format of the **OBSID1_mosaic[_bkg_sub].lblx** and **OBSID1_mosaic[_bkg_sub].img** files is very similar to the corresponding *reproj_img* files described earlier. The only difference is that the mosaic image files are always 18,000 × 401 since mosaic images are never clipped to include only valid data (see Section 2.5).

The **OBSID1_mosaic[_bkg_sub]_metadata_params.tab** file is also similar to the corresponding *reproj_img* file, with the exception that it contains two additional columns:

```
Corotating Longitude, Image Index, Mid-time SPICE ET, Inertial Longitude, Radial Resolution, Angular Resolution, Phase
Angle, Emission Angle
  0.00,   14, 296602880.000, 274.969,    5.27813,    0.00000,  37.882229,  93.791534
  0.02,   14, 296602880.000, 274.989,    5.28045,    0.00000,  37.882183,  93.791321
  0.04,   14, 296602880.000, 275.009,    5.28304,    0.00000,  37.882149,  93.791092
  0.06,   14, 296602880.000, 275.029,    5.28551,    0.00000,  37.882099,  93.790901
[...]
```

The first new column is "Image Index", which indicates which reprojected image was used to provide data for this particular co-rotating longitude. The second new column is "Mid-time SPICE ET", which gives the time in seconds corresponding to that source image. The number in the "Image Index" column references the file **OBSID1_mosaic[_bkg_sub]_metadata_src_imgs.tab**, which contains a list of the LIDVIDs of the reprojected images that were used to create this mosaic:

```
Source Image Index, LIDVID
   0, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022571n_reproj_img::1.0
   1, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022704n_reproj_img::1.0
   2, urn:nasa:pds:cassini_iss_fring_mosaics_rsfrench2025:data_reproj_img:1622022841n_reproj_img::1.0
[...]
```

3.2.4 browse_mosaic and *browse_mosaic_bkg_sub*

These directories contain browse images for the corresponding mosaics. Browse images are provided in four sizes:

- Thumb: 100×100 pixels
- Small: 200×200 pixels
- Medium: 1800×401 pixels
- Full: 18000×401 pixels

Each mosaic is downsampled as necessary to make the browse images. Each browse image is contrast-stretched to use the full available greyscale. Missing data in the mosaic is indicated by black.

3.3 The *document* directory

This directory contains the document collection and has the structure:

```
document/
├── collection_document.csv
├── collection_document.lblx
├── supplemental/
│   ├── global_mosaic_bkg_sub_index.lblx
│   ├── global_mosaic_bkg_sub_index.tab
│   ├── global_mosaic_index.lblx
│   ├── global_mosaic_index.tab
│   ├── global_reproj_img_index.lblx
│   └── global_reproj_img_index.tab
└── user_guide/
    ├── f-ring-mosaics-user-guide.lblx
    └── f-ring-mosaics-user-guide.pdf
```

3.3.1 user_guide

The *user_guide* directory contains the User's Guide (this document).

3.3.2 The global index files

This directory contains three global index files. Each index files contains summary information about each reprojected image or mosaic, as appropriate. The provided information is extensive and is designed to be useful to researchers who want to search for images or mosaics meeting certain requirements. Fields include:

- LID
- Observation ID
- File path
- Start and stop date/time and spacecraft clock count
- The number of co-rotating longitudes with valid data, and the percent coverage of the full 360° range
- The range of valid co-rotating and inertial longitudes
- The mean, minimum, and maximum phase angle, emission angle, radial resolution, and longitudinal resolution
- The mean incidence angle
- The minimum and maximum radius
- The product creation date
- Notes about the data product
- For a mosaic, the number of images that were used to create it, and the range of image names
- For a background-subtracted mosaic, that radial ranges used to compute the background model

For full details, see the index labels.

3.4 The *spice_kernels* directory

This directory contains the SPICE kernel collection, which consists of a single SPICE meta kernel listing all kernels used to process the data for this bundle (see Section 4.3).

4 External References and Further Reading

This section provides a curated list of external references, documentation, and further reading relevant to the Saturn F Ring Cassini ISS Mosaic Bundle.

4.1 PDS4 Standards and Documentation

- NASA Planetary Data System (PDS) Home: https://pds.nasa.gov/
- PDS4 Documents: https://pds.nasa.gov/datastandards/documents/
- PDS4 Information Model Specification: https://pds.nasa.gov/datastandards/documents/im/
- PDS4 Data Dictionaries: https://pds.nasa.gov/datastandards/dictionaries/

4.2 Cassini Mission and Instrumentation

- Cassini-Huygens Mission Overview: https://saturn.jpl.nasa.gov/mission/
- Cassini ISS (Imaging Science Subsystem) Overview: https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/Cassini/inst-iss.html
- Cassini ISS Data User Guide (PDS3): https://doi.org/10.17189/1504135
- Cassini ISS Narrow Angle Camera (NAC) Details: https://pds-rings.seti.org/pds4/bundles/cassini_iss_saturn/document/iss-na-camera-description.txt

- Cassini ISS Wide Angle Camera (WAC) Details: https://pds-rings.seti.org/pds4/bundles/cassini_iss_saturn/document/iss-wa-camera-description.txt

4.3 SPICE Toolkit

- SPICE Toolkit Documentation: https://naif.jpl.nasa.gov/naif/documentation.html
- SPICE Tutorials and Reference: https://naif.jpl.nasa.gov/naif/tutorials.html

4.4 Additional Resources

- PDS Ring-Moon Systems Node: https://pds-rings.seti.org/
- PDS Cartography and Imaging Sciences Node: https://pds-imaging.jpl.nasa.gov/

5 References

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